

# Report on Broad Gauge Locomotive

MECHANICAL ENGINEERING

TECHNICAL REPORT

BROAD GAUGE - 1676 MM



# Introduction & Fundamentals

Broad Gauge locomotives operate on a track gauge of **1,676 mm** and form the core of high-capacity rail networks across South Asia and select global systems. Their wider gauge enables superior stability, higher axle loads, greater hauling capacity, and elevated speed potential compared to narrower gauge alternatives.

## Operational Requirements

- High tractive effort for heavy loads
- Sustained hauling capability over long distances
- Speed compliance with network standards
- Rigorous safety and signaling integration
- High reliability and fleet availability

## Engineering Advantages

- Enhanced lateral stability at speed
- Higher permissible axle loads
- Increased freight and passenger capacity
- Reduced derailment risk
- Compatibility with modern signaling

# Classification of Broad Gauge Locomotives

## Electric Locomotives

Powered via overhead catenary (25 kV AC). High efficiency, low emissions, ideal for dense networks.

## Diesel-Electric Locomotives

Onboard diesel engine drives an alternator. Self-sufficient for non-electrified routes.

## Dual-Mode Locomotives

Operate on both electric and diesel power. Flexible for mixed infrastructure corridors.

## By Service Type

Passenger, Freight, Mixed-Traffic, and Shunting variants — each optimized for specific duty cycles.

# Structural Layout & Major Systems

A Broad Gauge locomotive integrates mechanical, electrical, and auxiliary subsystems within a robust carbody structure. The general arrangement includes the **carbody and underframe assembly**, driver cabs at both ends, equipment compartments housing power electronics, and two three-axle bogies providing guided wheel-rail contact and load distribution.



## Mechanical & Running Gear

- Wheelsets, axles, and primary/secondary suspension
- Two three-axle bogies per locomotive
- Carbody, underframe, and automatic coupler system



## Electrical & Power System

- **Electric:** Pantograph, Main Circuit Breaker, Transformer, Converter/Inverter
- **Diesel-Electric:** Diesel Engine, Main Alternator, Rectifier, Power Electronics



## Traction System

- Three-phase AC induction traction motors
- Gear drive and axle drive mechanism
- Precise tractive effort generation and control

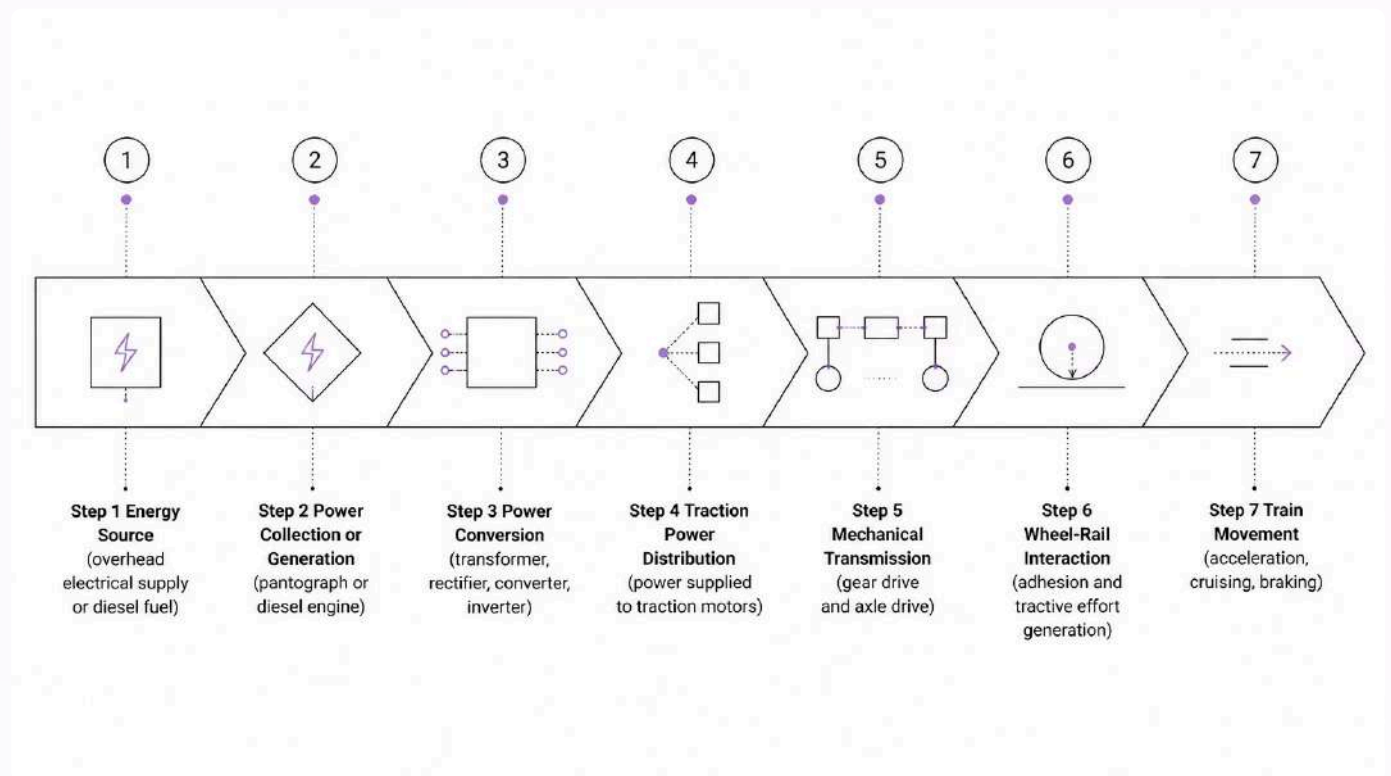


## Auxiliary Systems

- Air compressor for brake and pneumatic systems
- Cooling system, battery bank, and auxiliary converter
- Lighting, HVAC, and auxiliary load management

# Functional Process Flow & System Integration

The locomotive's operation follows a precisely sequenced energy-to-motion chain. From overhead electrical supply or onboard diesel fuel, power is collected, converted, distributed to traction motors, and transmitted mechanically to the wheel-rail interface — all governed by an advanced microprocessor-based control and monitoring architecture.



The complete functional flow diagram illustrates the sequential chain from energy acquisition through power conversion, traction distribution, mechanical transmission, and wheel-rail interaction — culminating in controlled train movement under continuous monitoring and protection.

## Control & Monitoring

- Driver Control Interface (DCI) with digital displays
- Microprocessor Control System (MPCS) for real-time management
- Sensors and feedback loops for speed, temperature, and current
- Onboard diagnostic systems for fault logging

## Safety & Protection

- Overcurrent and overvoltage protection circuits
- Wheel Slip Protection (WSP) with automatic correction
- Brake interlocks and emergency brake circuits
- Fire detection and suppression systems

# Locomotive Structure & Objectives

A broad gauge diesel-electric locomotive is organized into distinct compartments, each housing specialized equipment that contributes to the overall operation of the machine. Understanding this compartmental layout is essential for engineers, technicians, and operators who work with these powerful machines on a daily basis. The locomotive is typically divided into nine major compartments, each serving a critical function in the chain of energy conversion and power delivery.

## Nose Compartment

Houses battery bank, charging equipment, and auxiliary control systems for engine startup.

## Driver's Cabin

The command center with master controller, brake controls, and monitoring instruments.

## Electrical Compartment

Contains contactors, relays, circuit breakers, and control panels for power management.

## Generator Compartment

Houses the main generator that converts mechanical energy to electrical energy.

## Engine Compartment

Contains the prime mover — the diesel engine — along with turbocharger and fuel systems.

## Compressor Compartment

Produces compressed air for braking, horn operation, and pneumatic controls.

## Radiator Compartment

Manages thermal regulation through radiator core, cooling fans, and water pump.

## Front & Rear Bogies

Three-axle trucks supporting the locomotive weight and carrying traction motors.

The study of these locomotives is guided by four primary objectives: understanding the structural arrangement of locomotive compartments, studying the operation of major locomotive systems, analyzing the interaction between compartments and systems, and understanding maintenance and safety considerations. Together, these objectives form the foundation of locomotive engineering knowledge required for safe and efficient railway operations.

The major systems that govern locomotive operation include the Starting System, Power Transmission System, Air Intake System, Fuel Intake System, Water Cooling System, and Lubricating Oil System. Each of these systems operates in concert with the others to ensure reliable, continuous performance under varying load and environmental conditions. The starting system initiates engine operation, the air and fuel systems sustain combustion, the power transmission system delivers traction, and the cooling and lubrication systems protect the engine from wear and thermal damage.

# Nose Compartment & Starting System

The nose compartment forms the front section of the locomotive and houses essential auxiliary equipment required during locomotive startup and operation. This compartment is the first line of defense in ensuring that the locomotive can be brought to life reliably and safely. The equipment housed here operates at low voltage and provides the foundational electrical support needed before the main engine begins running. Without a properly functioning nose compartment, the entire locomotive would be unable to initiate its power generation cycle.

## Battery Bank

Provides DC electrical power for engine starting. The battery bank is the primary energy source used to crank the diesel engine during startup. It must maintain sufficient charge at all times to ensure reliable starting under all weather conditions.

## Battery Charging Equipment

Maintains battery voltage during operation. Once the engine is running, the charging equipment replenishes the battery, ensuring it is ready for the next startup cycle and for emergency power needs.

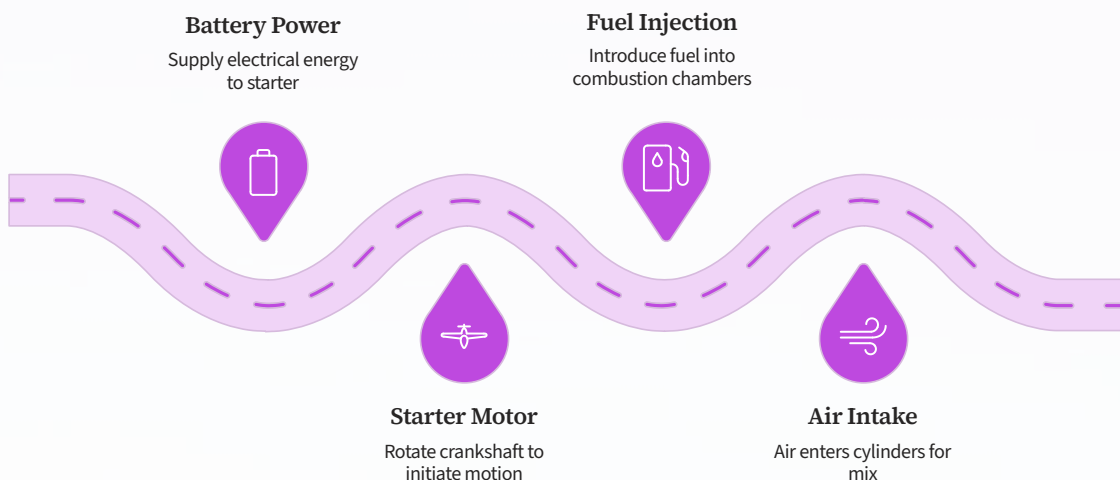
## Auxiliary Control Equipment

Supports low-voltage circuits and monitoring systems throughout the locomotive. This equipment ensures that control signals and instrumentation remain operational even when the main power system is not yet active.

## Emergency Equipment

Includes safety and emergency operation devices that can be activated independently of the main systems. This equipment is critical for safe shutdown procedures and emergency response scenarios.

The starting system initiates engine operation by supplying electrical energy to the starter mechanism. The process follows a well-defined sequence: the battery supplies electrical power, the starter motor rotates the crankshaft, fuel injection begins, air enters the cylinders, combustion starts, and finally the engine runs independently. This six-step process must occur within seconds to ensure a smooth and reliable engine start. The system includes batteries, starting contactors, a starter motor, control relays, and protection circuits that work together to manage the high current demands of engine cranking while protecting the electrical system from damage.



The reliability of the starting system is paramount for locomotive operations. A failed start can result in significant delays to train schedules, increased operational costs, and potential safety hazards if the locomotive is stranded on an active line. Regular inspection of batteries, contactors, and starter motors is therefore a critical part of daily maintenance routines.



**NOSE COMPARTMENT**



**BATTERY BANK**



# Driver's Cabin & Operational Controls

The driver's cabin serves as the nerve center of the locomotive, where all operational decisions are made and all system performance is monitored. It is designed to provide the locomotive engineer with complete control over the machine while ensuring maximum safety and situational awareness. The cabin is ergonomically arranged to minimize operator fatigue during long hauls and is equipped with redundant safety systems to protect both the crew and the train.

## Master Controller

The master controller is the primary interface for controlling locomotive speed and power output. It allows the driver to select power notches that regulate the amount of electrical power delivered to the traction motors, enabling precise speed control under varying track conditions and train loads.

## Brake Control System

The brake control system manages train braking through a combination of dynamic braking and air brake application. The driver can modulate braking force to ensure smooth deceleration and safe stopping distances.

## Monitoring Instruments

- **Speedometer** — Displays current locomotive speed in km/h
- **Pressure Gauges** — Monitor brake pipe, main reservoir, and cylinder pressures
- **Temperature Indicators** — Track engine coolant, oil, and generator temperatures
- **Electrical Meters** — Display voltage, current, and power output readings

## Communication Equipment

- Radio communication systems for coordination with control centers
- Warning systems including horn and bell controls
- Emergency alert devices for crew communication

The driver's cabin is also equipped with a comprehensive suite of safety features designed to protect against human error and equipment failure. The dead-man switch is one of the most critical safety devices — it requires the driver to maintain continuous pressure on a control, and if released (indicating the driver may be incapacitated), it automatically applies the emergency brakes. Emergency brake application buttons are strategically placed for rapid activation. Alarm systems provide audible and visual warnings for abnormal operating conditions, while fire warning indicators alert the crew to potential thermal hazards in the engine and electrical compartments.



### Dead-Man Switch

Requires continuous driver input; triggers emergency brakes if released.



### Emergency Brake

Rapid full-service brake application for immediate train stopping.



### Alarm Systems

Audible and visual alerts for abnormal operating conditions.



### Fire Warning

Detects thermal anomalies in engine and electrical compartments.



**DRIVER'S CABIN**



**THROTTLE HANDLE,  
BRAKE CONTROLS &  
HORN CONTROLS**



**SPEEDOMETER,  
VIGILANCE  
CONTROLS**

# Electrical Compartment & Power Transmission System

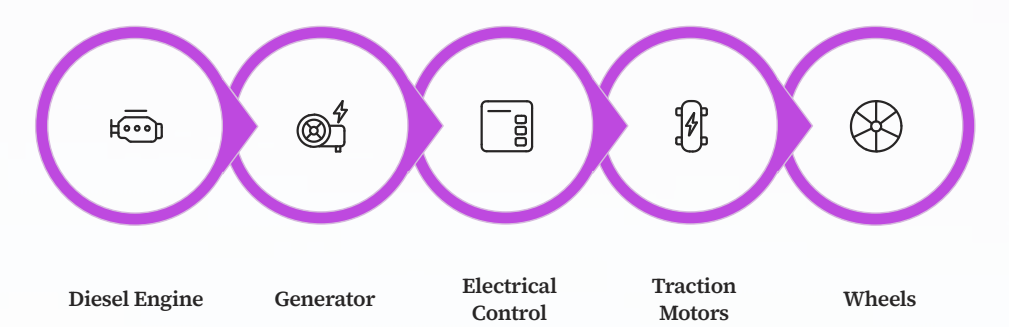
The electrical compartment contains the control and protection devices responsible for managing electrical power flow throughout the locomotive. This compartment acts as the nervous system of the machine, ensuring that the electrical energy generated by the main generator is properly conditioned, protected, and distributed to the traction motors and auxiliary systems. Without the electrical compartment's sophisticated control equipment, the raw electrical output of the generator could not be safely or efficiently utilized for traction purposes.

## Major Components

|                                                                                                                                                                                   |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contactors</b><br>Heavy-duty electromagnetic switches that control high-power circuits, enabling or interrupting power flow to traction motors and other major loads.          | <b>Relays</b><br>Provide automatic switching based on electrical conditions, enabling protective functions and control logic without manual intervention.             |
| <b>Circuit Breakers</b><br>Protect against electrical faults by interrupting current flow when overcurrent or short-circuit conditions are detected, preventing equipment damage. | <b>Control Panels</b><br>Centralized monitoring interfaces that display system status, fault codes, and operational parameters to the crew and maintenance personnel. |

## Power Transmission Flow

The power transmission system is responsible for transferring generated electrical power from the main generator to the traction motors mounted on the bogies. This system must handle extremely high currents and voltages while maintaining precise control over motor speed and torque. The flow of power follows a defined path through several stages of conditioning and control before reaching the wheels.



The power transmission system performs four essential functions: power regulation ensures that the correct amount of electrical energy is delivered to the traction motors at all times; load control manages the distribution of power across multiple motors to prevent wheel slip and ensure even traction; motor protection safeguards the traction motors from electrical faults, overheating, and mechanical overload; and efficient traction control optimizes the relationship between speed, torque, and power to maximize locomotive performance under varying track gradients and train loads. Together, these functions enable the diesel-electric locomotive to haul heavy freight and passenger trains with precision and reliability.



**ELECTRICAL  
CIRCUITS**



**CONTROL PANELS**



# Generator Compartment

The generator compartment houses the main generator or alternator — the heart of the diesel-electric power conversion system. This is where the mechanical energy produced by the diesel engine is transformed into the electrical energy that ultimately drives the traction motors. The generator is directly coupled to the engine crankshaft and rotates at the same speed, producing alternating current (AC) that is subsequently rectified and controlled before being supplied to the traction motors. The efficiency and reliability of the generator directly determine the overall performance of the locomotive.



## Rotor

The rotor is the rotating magnetic element of the generator. It is driven directly by the diesel engine and creates a rotating magnetic field that induces electrical current in the stator windings. The rotor must be precisely balanced to operate smoothly at high rotational speeds.



## Excitation System

The excitation system controls voltage generation by regulating the magnetic field strength in the rotor. By adjusting the excitation current, the system can control the output voltage and power of the generator to match the demands of the traction motors under varying load conditions.



## Stator

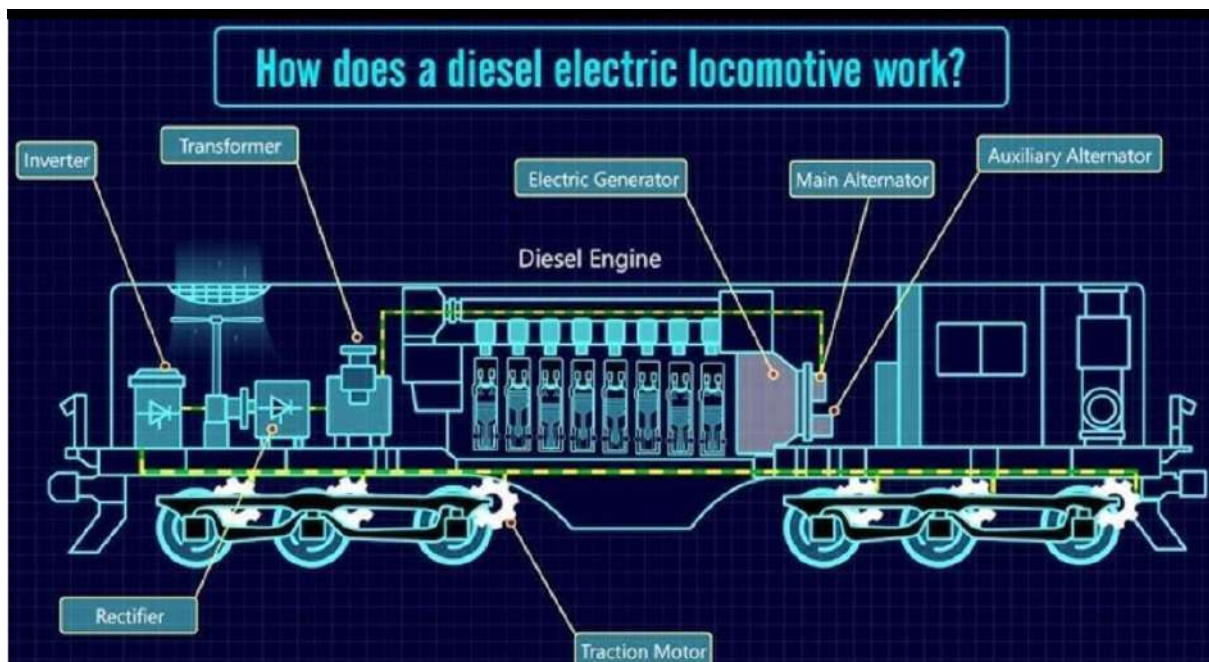
The stator is the stationary part of the generator that surrounds the rotor. It contains the windings in which electrical current is induced by the rotating magnetic field. The stator produces the electrical output that is fed to the traction motors through the electrical control equipment.



## Cooling Equipment

The generator produces significant heat during operation due to electrical resistance and magnetic losses. Cooling equipment, including forced-air ventilation and sometimes liquid cooling, maintains the generator within its safe operating temperature range to prevent insulation damage and ensure long service life.

The energy conversion process in a diesel-electric locomotive follows a three-stage transformation: chemical energy stored in diesel fuel is converted to mechanical energy through combustion in the engine cylinders, and this mechanical energy is then converted to electrical energy by the generator. This indirect method of power transmission offers significant advantages over direct mechanical drive, including smooth torque delivery, precise speed control, and the ability to distribute power to multiple axles simultaneously without complex mechanical linkages. The generator compartment is therefore central to the operational philosophy of the diesel-electric locomotive, enabling the high efficiency and reliability that make these machines the backbone of modern rail freight and passenger transport.



Main generator (Alternator)



**Multi-groove belt pulley (sheave), and the belts are V-belts transmitting power between rotating components.**

# Engine Compartment, Air Intake & Fuel System

The engine compartment houses the prime mover of the locomotive — a large, multi-cylinder diesel engine that serves as the primary source of mechanical power. This engine is typically a two-stroke or four-stroke design with multiple cylinders arranged in a V-configuration, capable of producing thousands of horsepower. The engine compartment is the largest and most complex section of the locomotive, containing not only the engine itself but also the associated systems that supply it with air and fuel, and manage its thermal output.

## Main Engine Components

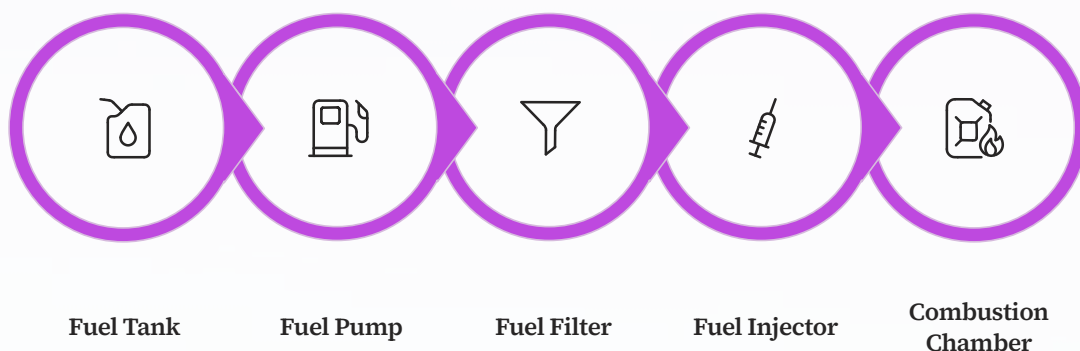
- **Cylinder Block** — The structural foundation that supports all engine cylinders and contains the coolant passages
- **Pistons** — Convert combustion pressure into reciprocating linear motion
- **Crankshaft** — Converts reciprocating piston motion into rotary motion to drive the generator
- **Turbocharger** — Uses exhaust gas energy to compress intake air, increasing engine power output

## Air Intake System

Supplies clean, filtered air for combustion. Atmospheric air passes through the air filter, is compressed by the turbocharger, and enters the engine cylinders through the intake manifold. This system improves combustion efficiency, increases power output, and reduces emissions by ensuring the correct air-fuel ratio.

## Fuel Intake System

The fuel intake system is responsible for storing, filtering, pressurizing, and delivering diesel fuel to the engine cylinders with precise timing and quantity. The system begins at the fuel tank, which stores the locomotive's diesel fuel supply. A fuel pump draws fuel from the tank and pressurizes it for delivery through the fuel filters, which remove contaminants and water that could damage the precision fuel injectors. The filtered fuel is then delivered to the fuel injectors, which spray a fine mist of fuel into the combustion chamber at the precise moment required for efficient combustion.



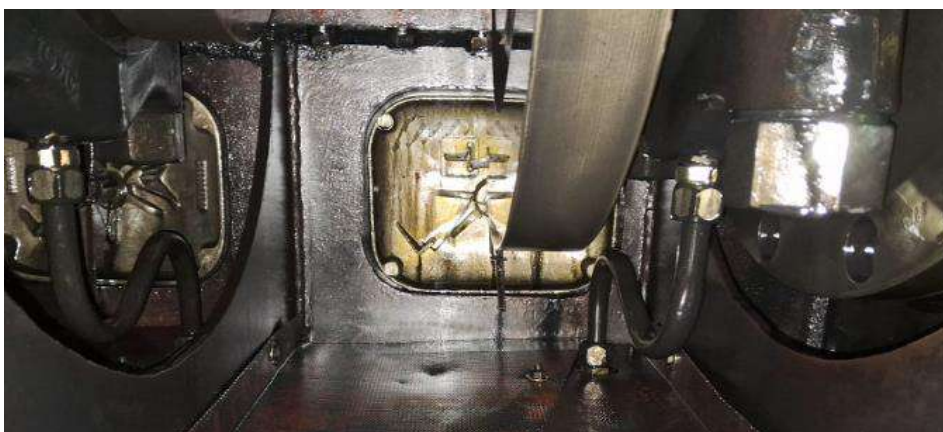
The fuel injectors are among the most critical components in the fuel system. They must deliver fuel at extremely high pressures — often exceeding 20,000 psi — to ensure proper atomization and complete combustion. The timing and duration of fuel injection are precisely controlled to optimize engine performance, fuel economy, and emissions. Any degradation in fuel quality, filter condition, or injector performance can result in reduced power output, increased fuel consumption, and accelerated engine wear. Regular maintenance of the fuel system, including filter replacement and injector testing, is therefore essential for maintaining locomotive reliability and efficiency.



**Engine Block** – The main body of the diesel engine. It houses the cylinders, pistons, liners, crankshaft supports, and coolant passages.



**Cylinder Head (Head)** – The top section of the engine, bolted above the cylinders. It contains the intake and exhaust valves, injector, and combustion chamber components.



**Sump (Oil Pan/Crankcase Sump)** – The lowest part of the engine, beneath the engine block. It stores the lubricating oil that is circulated throughout the engine.

# Engine Combustion Process & Performance

The combustion process within a diesel engine is a precisely timed sequence of events that converts the chemical energy of fuel into mechanical work. Understanding this process is fundamental to understanding how the locomotive generates power and what factors influence its performance, efficiency, and emissions. The diesel combustion cycle differs from gasoline engines in that ignition occurs through compression heating rather than spark ignition, making the diesel engine inherently more thermally efficient.

**1**

## Air Compression

Air enters the cylinder and is compressed by the rising piston, raising its temperature to approximately 500–700°C.

**2**

## Fuel Injection

Fuel is injected at high pressure into the hot compressed air, atomizing into fine droplets for rapid mixing.

**3**

## Ignition

Fuel ignites spontaneously due to the high temperature of the compressed air — no spark plug required.

**4**

## Expansion Stroke

Combustion pressure forces the piston downward, generating mechanical power transmitted to the crankshaft.

**5**

## Exhaust Stroke

Burnt gases are expelled from the cylinder through the exhaust valve, completing the cycle.

## Factors Affecting Performance

Engine performance is influenced by a combination of operational and maintenance factors. Air quality affects combustion efficiency — dirty or humid air reduces power output and increases wear. Fuel quality determines combustion completeness and deposit formation. Lubrication condition affects friction losses and component wear. Cooling efficiency prevents thermal degradation of engine components. Regular engine maintenance ensures all systems operate within design parameters and extends service life.

## Engine Protection Systems

### Over-Speed Protection

Prevents engine damage from excessive RPM by cutting fuel supply.

### Low Oil Pressure

Shuts down engine if lubrication pressure falls below safe threshold.

### High-Temperature

Triggers alarms and derating if coolant or oil temperature exceeds limits.

# Compressor, Radiator, Cooling & Lubrication Systems

The compressor compartment and radiator compartment house the systems that support the engine's pneumatic and thermal management requirements. These systems operate continuously during locomotive operation and are essential for maintaining safe and efficient performance. The compressor provides the compressed air needed for braking and pneumatic controls, while the radiator and water cooling system manage the substantial heat generated by the diesel engine during combustion.

## Compressor Compartment

The air compressor produces compressed air that is stored in main reservoir tanks and distributed throughout the locomotive and train. This compressed air is used for three primary purposes: air brakes (the primary braking system for the locomotive and all cars in the train), horn operation (the locomotive's audible warning device), and pneumatic controls (various actuators and valves throughout the locomotive). The compressor is typically driven by the main engine and must maintain adequate reservoir pressure at all times to ensure brake system availability.

## Radiator Compartment

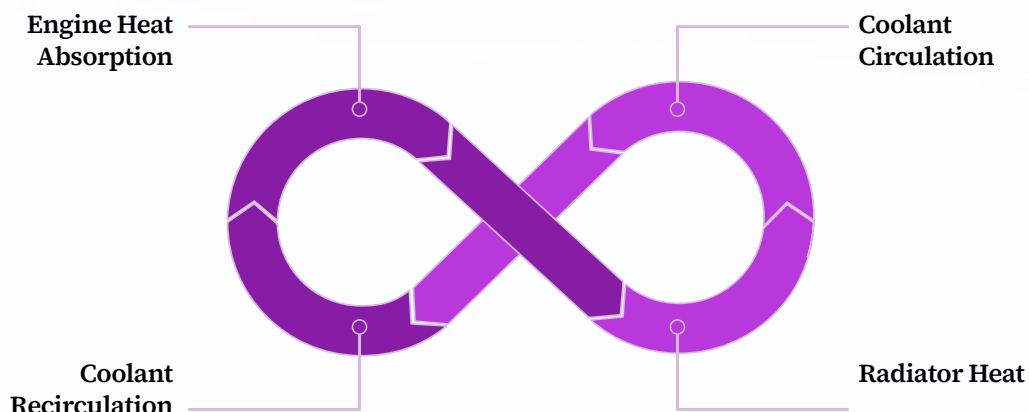
The radiator compartment removes excess heat generated by engine operation. It contains the radiator core, through which hot coolant flows; cooling fans that draw ambient air through the radiator; a water pump that circulates coolant; and a thermostat that regulates coolant flow to maintain optimal operating temperature. The radiator compartment is typically located at the rear of the locomotive and is designed for maximum airflow efficiency.

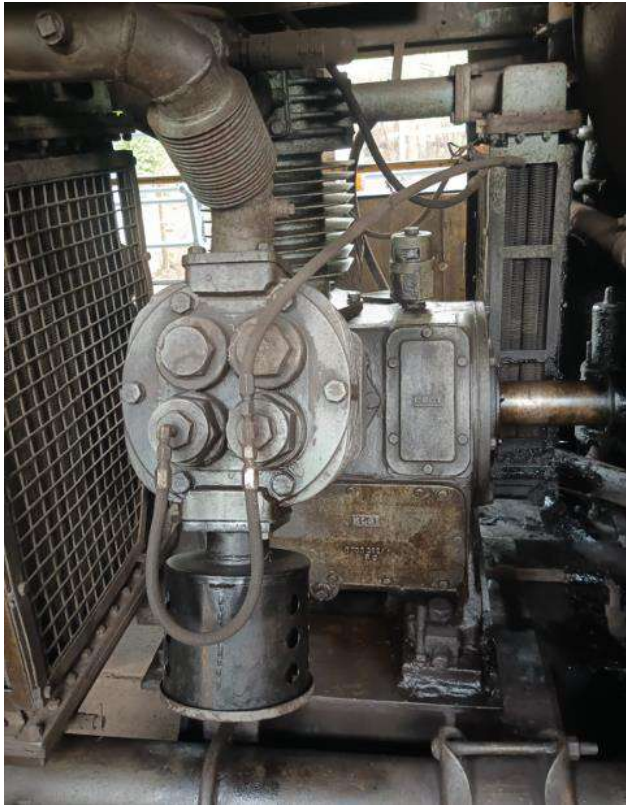
## Water Cooling System

The water cooling system circulates coolant through the engine block, absorbing heat from the cylinders and cylinder heads. The heated coolant then flows to the radiator, where heat is dissipated to the atmosphere before the coolant is recirculated. This continuous loop prevents overheating, maintains optimum operating temperature, and significantly improves engine life by reducing thermal stress on components.

## Lubricating Oil System

The lubricating oil system reduces friction between moving engine components, preventing wear and extending service life. Oil is drawn from the reservoir by the oil pump, circulated through the engine bearings, pistons, and crankshaft, then passed through the oil filter to remove contaminants and the oil cooler to remove heat before returning to the reservoir. This system is critical for engine longevity and efficiency.





## Reciprocating Air Compressor

Reciprocating air compressor assembly and functions to transmit rotational motion from the drive system to the compressor mechanism. It plays a vital role in the compression process by supporting the movement required to draw in, compress, and discharge air efficiently.



The impeller is the rotating component of a centrifugal pump that imparts kinetic energy to the fluid, increasing its velocity and pressure. Its curved vanes direct the fluid from the center to the outer edge, enabling efficient fluid transfer through the pumping system.

## FRONT TRUCK / BOGIE / TROLLEY



- Bogie Frame
- Wheelset (Wheels & Axle)
- Axle Box Bearing
- Suspension Springs
- Traction Motor
- Gear Unit
- Brake Cylinder
- Brake Rigging
- Center Pivot
- Side Bearers

The bogie wheelset assembly supports the weight of the locomotive and provides guidance along the railway track. It transmits traction and braking forces between the locomotive and rails, ensuring stable movement, load distribution, and safe operation.

### REAR TRUCK-

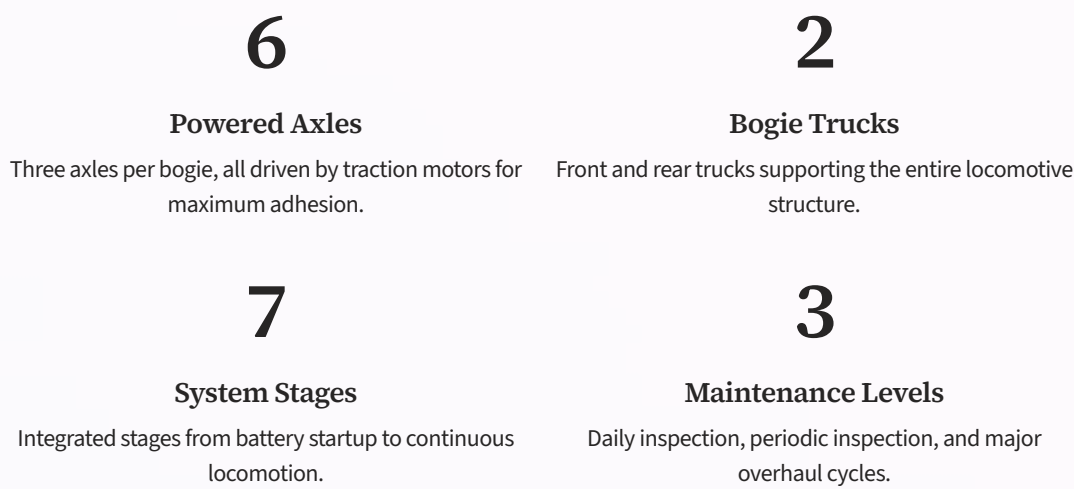


The rear truck assembly is a critical running gear component that supports the locomotive body and ensures safe movement along the rails. It facilitates load distribution, track guidance, and the transmission of braking and traction forces.

**The Over Speed Trip Device (OSTD)** is a safety device fitted in diesel locomotives to prevent the engine from exceeding its permissible speed limit. If the locomotive speed goes beyond the set safe limit, the device automatically trips the engine or cuts off fuel supply to avoid damage and ensure safe operation. It protects both the locomotive and passengers by preventing overspeed conditions.

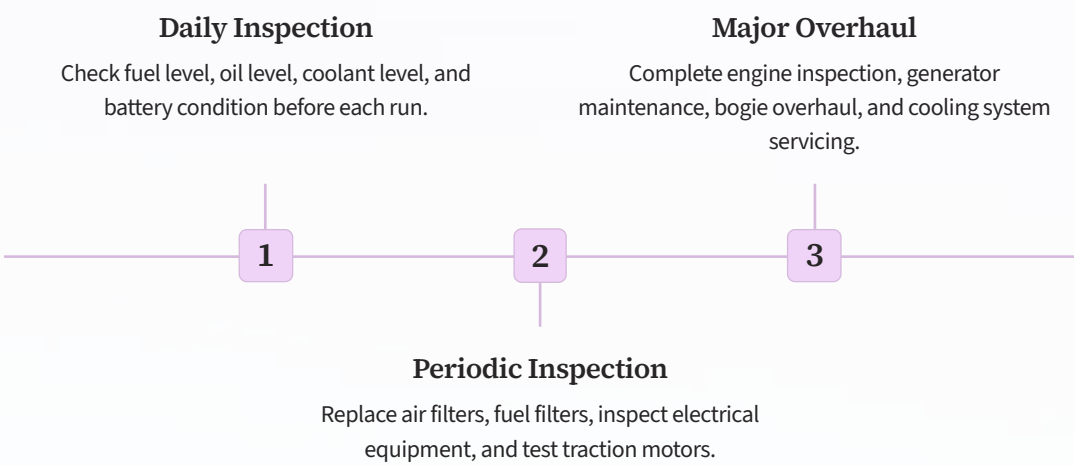
# Bogies, System Integration & Maintenance

The front and rear trucks (bogies) are the structural foundation upon which the entire locomotive rests. Each bogie typically consists of three axles, giving the locomotive a total of six powered axles. The bogies support the locomotive's weight, carry the traction motors, maintain track stability, and absorb shocks and vibrations from the rail. They are connected to the locomotive body through a pivot arrangement that allows them to rotate independently, enabling the locomotive to negotiate curves smoothly. The traction motors mounted on the bogies receive electrical power from the generator and convert it into the rotational force that drives the wheels.



The complete diesel-electric locomotive operates as an integrated system where all compartments and subsystems work in concert. The sequence begins with the battery initiating engine startup, followed by simultaneous operation of the air intake and fuel intake systems. The diesel engine generates mechanical power through combustion, which the generator converts into electricity. This electricity powers the traction motors through the electrical control equipment, and the traction motors rotate the wheels through the bogies to produce locomotion. Throughout this entire process, the water cooling system, lubricating oil system, and air compressor system operate continuously to support and protect the engine and auxiliary equipment.

## Maintenance Practices



The diesel-electric locomotive is a complex integration of mechanical, electrical, thermal, pneumatic, and control systems. Each compartment performs a specialized role while contributing to overall locomotive performance. The starting system initiates operation, the air and fuel systems sustain combustion, the generator and electrical systems transmit power, and the cooling and lubrication systems ensure reliability and longevity. Together with the bogies and traction equipment, these systems enable safe, efficient, and continuous railway transportation.

OFFICIAL TECHNICAL REPORT

# VPA Sewage Treatment Plant: Operational Excellence Report

VIZAG  
PORT

An aerial photograph of the Vizag Port, showing a large container ship docked at a quay. The ship's deck is covered with numerous colorful shipping containers. In the background, the port yard is filled with stacks of containers and various port infrastructure. The city of Vizag is visible in the distance, along with the Bay of Bengal.

# Introduction to VPA Sewage Treatment Plant

## 1.1 About Visakhapatnam Port Authority

Visakhapatnam Port Authority (VPA), also known as Vizag Port, is one of India's largest major ports located on the eastern coast along the Bay of Bengal. As a critical maritime hub handling cargo, industrial effluents, and supporting a large workforce, VPA generates significant volumes of domestic and industrial sewage requiring systematic treatment before discharge.

## 1.2 Need for Sewage Treatment

Untreated sewage discharged into marine environments causes severe ecological damage – depleting dissolved oxygen, spreading waterborne diseases, and destroying marine biodiversity. VPA's STP ensures all wastewater is treated to **CPCB (Central Pollution Control Board)** standards before any discharge or reuse.

## 1.3 Sources of Sewage in VPA

- & Residential colonies within the port township
- & Administrative and office buildings
- & Port hospital and medical facilities
- & Maintenance workshops and garages
- & Staff canteens and food service areas

## 1.4 Objectives of STP

- Treat all domestic sewage generated within VPA premises
- Achieve discharge standards as per CPCB norms
- Enable treated water reuse for non-potable applications
- Protect the Bay of Bengal marine ecosystem
- Support sustainable port operations

## 1.5 Plant Capacity

10MLD

Design Capacity

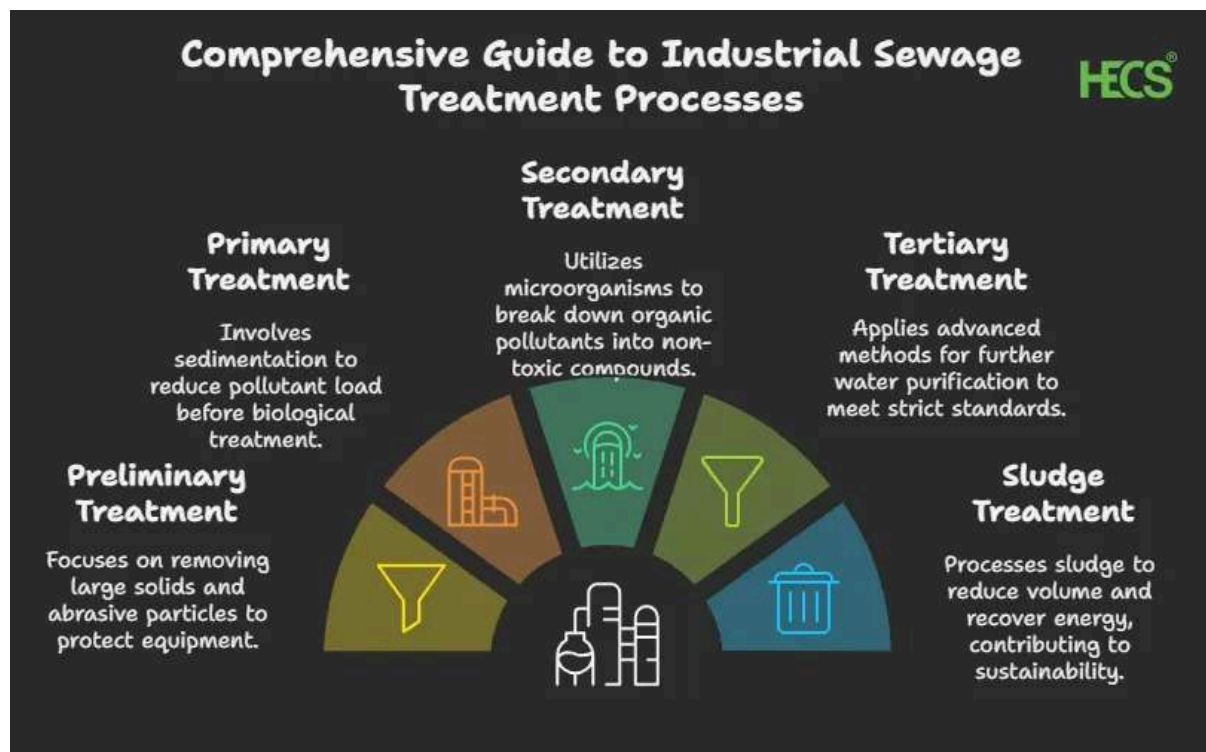
Million Litres per Day

300

BOD Inlet (mg/L)

Average raw sewage BOD

- ✔ **10 MLD Sewage Treatment Capacity** – VPA STP is designed to handle the complete wastewater load of the port authority's township and facilities.



A **Sewage Treatment Plant (STP)** is a facility used to remove contaminants, impurities, and harmful microorganisms from sewage (wastewater) before releasing it into rivers, lakes, or for reuse.

### Objectives of STP

1. Remove suspended solids.
2. Remove organic matter.
3. Reduce harmful bacteria and pathogens.
4. Prevent water pollution.
5. Make water suitable for reuse or safe disposal.

# Complete Process Flow of STP

The VPA STP operates through a sequential multi-stage treatment process. Each stage targets specific pollutants, progressively purifying the sewage until it meets discharge and reuse standards. The hydraulic flow is gravity-driven wherever possible, with pumping employed at designated lift stations.



The above diagram illustrates the complete hydraulic pathway from raw sewage inlet to final treated water output, highlighting each critical purification stage within the VPA STP.

## 2.2 Treatment Stages Overview

|                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------|
| 01                                                                                                                          |
| <b>Preliminary Treatment</b><br><br>Screening and grit removal eliminate large solids and abrasive particles.               |
| 02                                                                                                                          |
| <b>Primary Treatment</b><br><br>Sedimentation in clarifier removes settleable suspended solids.                             |
| 03                                                                                                                          |
| <b>Secondary (Biological) Treatment</b><br><br>Activated sludge process with aeration degrades organic matter biologically. |
| 04                                                                                                                          |
| <b>Tertiary Treatment &amp; Disinfection</b><br><br>Chlorination eliminates pathogens; treated water stored for reuse.      |

## 2.4 Pollutants Removed at Each Stage

| Stage               | Pollutant Removed            |
|---------------------|------------------------------|
| Bar Screening       | Plastics, cloth, rags, wood  |
| Grit Chamber        | Sand, gravel, eggshells      |
| Primary Clarifier   | Suspended solids, scum       |
| Aeration Tank       | BOD, COD, organics           |
| Secondary Clarifier | Biological sludge            |
| Chlorination        | Bacteria, viruses, parasites |

# Preliminary Treatment System

Preliminary treatment is the first line of defence in the STP. It protects downstream mechanical equipment from damage caused by large debris and abrasive particles. This stage includes three key units: the Bar Screen Chamber, Grit Chamber, and Collection Sump.



## 3.1 Bar Screen Chamber

**Components:** Screen bars (6–20mm spacing), steel frame, mechanical rake mechanism, drive motor with gearbox.

**Function:** Removes plastics, cloth, rags, wood pieces, and other large debris from incoming sewage. The rake mechanism automatically lifts trapped material onto a conveyor for disposal.

**Working Principle:** Sewage flows through inclined screen bars; solids are retained and periodically removed by the motorised rake.

3.2

## Grit Chamber

**Components:** Grit tank (channel type), hopper bottom, chain-and-flight scraper or air-induced vortex system.

**Function:** Removes sand, gravel, eggshells, and other heavy inorganic particles. Protects pumps and mechanical equipment from abrasion and wear.

**Working Principle:** Flow velocity is reduced to 0.3 m/s, allowing heavy grit to settle while organic matter remains suspended.

3.3



## Collection Sump (Wet Well)

**Components:** Wet well chamber, float switches (level sensors), submersible sewage pump intake.

**Function:** Provides flow equalisation and temporary storage. Ensures steady feed to primary treatment regardless of inflow fluctuations.

**Working Principle:** Float switches activate pumps at high level and stop them at low level, maintaining controlled flow to downstream units.

# Pumping System & Primary Clarifier

## 4.1 Sewage Pumping System

Submersible sewage pumps transfer screened sewage from the collection sump to the primary clarifier and subsequent treatment units. These are non-clog pumps designed to handle solids-laden wastewater.

### Key Components:

- & **Impeller:** Vortex or channel type, handles solids up to 50mm
- & **Shaft s Bearings:** Stainless steel shaft with grease-lubricated bearings
- & **Mechanical Seal:** Double mechanical seal prevents leakage into motor
- & **Drive Motor:** TEFC (Totally Enclosed Fan Cooled), 3-phase induction motor

 Pumps are typically sized for 10–15 MLD capacity with standby provision for redundancy.

## 4.2 Primary Clarifier

The primary clarifier is a circular or rectangular sedimentation tank where sewage is held at low velocity, allowing heavy suspended solids to settle by gravity.

### Components:

- ◆ **Circular Tank:** RCC structure, 3–4m depth
- ◆ **Scraper Arm:** Rotating bridge with scraper blades
- ◆ **Sludge Hopper:** Conical bottom for sludge collection
- ◆ **Surface Skimmer:** Removes floating scum and grease

### Removal Efficiency:

50–60%

Suspended Solids (TSS)

30–40%

BOD Reduction

| DELTA PUMP HOUSE |           |            |            |                                 |               |    |      |     |
|------------------|-----------|------------|------------|---------------------------------|---------------|----|------|-----|
| PUMP DETAILS     |           |            |            |                                 | MOTOR DETAILS |    |      |     |
| S.L. No.         | MAKE      | TYPE       | TOTAL HEAD | DISCHARGE<br>M <sup>3</sup> /Hr | SPEED<br>RPM  | HP | AMPS | RPM |
| PUMP No.1        | KIRLOSKAR | MF 25 / 25 | 8 Mtrs     | 420/Hr                          | 940           | 20 | 28   | 980 |
| PUMP No.2        | "         | "          | "          | "                               | "             | "  | "    | "   |
| PUMP No.3        | "         | "          | "          | "                               | "             | "  | "    | "   |

| RAW SEWAGE PUMPS 4 No's |      |                  |
|-------------------------|------|------------------|
| RSP No-1                | 30HP | Mud Pump         |
| RSP No-2                | 30HP | Submersible Pump |
| RSP No-3                | 20HP | Vertical Pump    |
| RSP No-4                | 30HP | Mud Pump         |

# Aeration Tank – Heart of the Sewage Treatment Plant

- ✔ **Core Purification Stage:** The aeration tank is where 85–90% of organic pollutant removal occurs through biological oxidation – the most critical purification process in the entire STP.

## 5.1 Activated Sludge Process

The Extended Aeration Activated Sludge Process is employed at VPA STP.

Microorganisms (activated sludge) consume organic matter in the presence of dissolved oxygen, converting BOD into  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , and new cell mass.

## 5.3 Role of Microorganisms

**Aerobic Bacteria:** Degrade carbonaceous organic matter (BOD). **Nitrifying Bacteria**

**(Nitrosomonas s Nitrobacter):** Convert ammonia to nitrate (nitrification). **Floc-forming**

**Bacteria:** Create settleable biomass for secondary clarification.

## 5.4 Aeration Tank Mechanical Components

**Roots Blowers / Air Blowers:** Supply continuous pressurised air (60–80 kPa). **Fine Bubble Diffusers:** EPDM membrane diffusers release micro-bubbles for maximum oxygen transfer efficiency (OTE 25–35%). **Air Header Pipes:** MS/SS piping distributes air uniformly. **Isolation Valves s Flow Meters:** Control and monitor air flow to each zone.

## 5.5–5.7 Oxygen Transfer, BOD & COD Removal

Air is injected through diffusers at the tank bottom. Rising bubbles transfer oxygen into the mixed liquor (MLSS 2000–4000 mg/L). Aerobic bacteria oxidise organics, achieving **BOD removal of 85–95%** and **COD reduction of 80–90%**. Typical HRT (Hydraulic Retention Time): 12–24 hours.





## Aeration Machine Parts and Functions

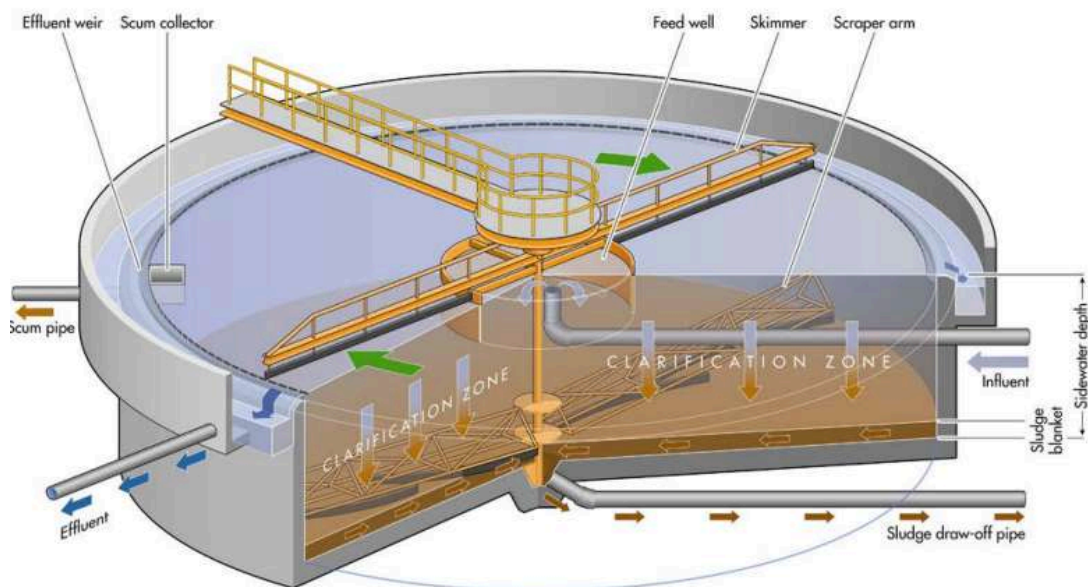
- **Motor:** Drives the aerator by converting electrical energy into rotational motion.
- **Gearbox:** Reduces speed and increases torque for efficient

operation.

- **Coupling:** Connects the motor and gearbox shafts for power transmission.
- **Shaft:** Transfers rotational power from the gearbox to the impeller.
- **Impeller/Fan:** Mixes wastewater and introduces oxygen into it.
- **Bearings:** Support the rotating shaft and reduce friction.
- **Protective Cover:** Protects equipment from dust, moisture, and damage.
- **Mounting Frame:** Holds and supports all aerator components in position.

### Working:

The motor rotates the gearbox, which drives the shaft and impeller. The impeller agitates the wastewater, allowing oxygen from the atmosphere to dissolve into the water. This oxygen helps microorganisms break down organic pollutants during the treatment process.



**Figure 2 – Circular Clarifier**

# Secondary Clarifier & Sludge Recirculation

## 6.1 Secondary Clarifier

Following biological treatment in the aeration tank, the mixed liquor (treated sewage + activated sludge) flows to the secondary clarifier. Here, biological sludge flocs settle by gravity, separating clean treated water from the biomass.

### Components:

- & **Clarifier Tank:** Circular RCC tank, similar design to primary clarifier
- & **Rotating Scraper:** Slow-speed scraper (0.3–1 rpm) collects settled sludge
- & **Sludge Hopper:** Central conical hopper for sludge collection
- & **Effluent Weir:** V-notch weir collects clarified supernatant

## 6.2 Settling Process

Surface overflow rate is maintained at 15–30 m<sup>3</sup>/m<sup>2</sup>/day. Settling time is typically 2–4 hours. Effluent turbidity should be <5 NTU before chlorination.

## 6.3 Return Activated Sludge (RAS)

A portion of settled sludge (20–40% of influent flow) is pumped back to the aeration tank inlet to maintain adequate MLSS concentration and bacterial population for continuous biological treatment.

## 6.4 Waste Activated Sludge (WAS)

Excess sludge generated due to microbial growth is removed as WAS and sent to the sludge treatment system.

WAS rate is controlled to maintain Sludge Age (SRT) of 15–30 days in extended aeration systems.

### RAS Flow

20–40% of Q

### WAS Flow



# Disinfection, Treated Water Storage & Sludge Treatment

## 7.1 Chlorination System

Chlorination is the final barrier against pathogenic microorganisms. Treated effluent from the secondary clarifier is dosed with chlorine before storage.

### Components:

- **Chlorine Tank:** HDPE/FRP tank storing liquid chlorine or sodium hypochlorite
- **Dosing Pump:** Metering pump for precise chlorine dosage (5–10 mg/L)
- **Contact Tank:** Provides 30-minute contact time for effective disinfection

**Pathogens Eliminated:** Coliform bacteria, E. coli, viruses, Giardia, Cryptosporidium.

## 7.4 Treated Water Storage & Reuse

Treated and disinfected water is stored in RCC/HDPE tanks and reused for:

- Road sprinkling within port premises
- Dust suppression at cargo handling areas
- Green belt and garden irrigation
- Industrial cooling (where applicable)

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## Sludge Moisture Reduction Table

| Stage            | Moisture Content | Solids % |
|------------------|------------------|----------|
| Raw Sludge       | 95%              | 5%       |
| Thickened Sludge | 75%              | 25%      |
| Dried Sludge     | 20%              | 80%      |

## –8.4 Sludge Treatment System

Sludge from primary clarifier (primary sludge) and WAS from secondary clarifier (biological sludge) are combined and processed through the sludge treatment train.

### 1 Sludge Thickener

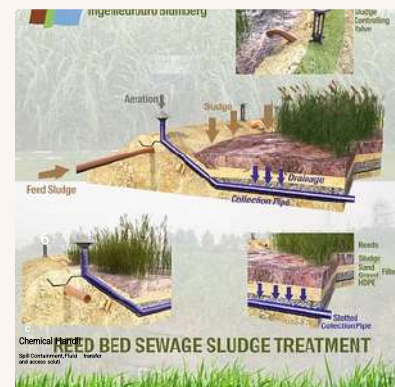
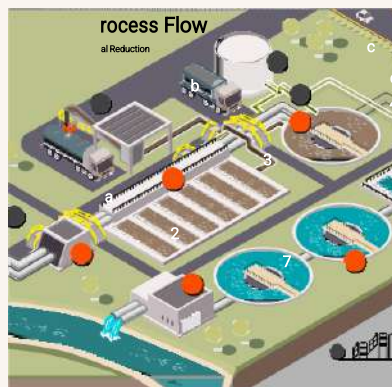
Gravity thickener increases solids from 1% to 3–5%. Components: thickener tank, slow-speed scraper, sludge hopper.

### 2 Sludge Drying Beds

Sand-gravel layered beds with under-drain system. Sun drying reduces moisture to 20%. Dried sludge disposed at authorised landfill or used for land application.

### Most Common Chemical in STPs

**Sodium Hypochlorite (NaOCl)** and **Bleaching power (CaOCl<sub>2</sub>)** are the most commonly used chemicals for chlorination in modern sewage treatment plants.



# "Pumping System of Sewage Treatment Plant (STP)

## Purpose

Pumping system transfers sewage, sludge, and treated water between STP units.

### Main Pump House

- . 3 Pumps × 60 HP
- . Receives sewage from collection sump
- . Pumps wastewater to treatment units
- . Ensures continuous plant operation

### New MDSS Pump House

- . 2 Pumps × 60 HP
- . Transfers sewage between treatment stages
- . Operates under PLC control

### VPA Pump House

- . 30 HP Pumps
- . Distributes sewage and treated water
- . Supports utility and reuse operations

### Primary Pumps

- . Lift incoming sewage
- . Maintain required flow rate
- . Handle wastewater with suspended solids

### Odour Control System

- . Reduces sewage foul smell
- . Improves environmental conditions
- . Enhances workplace safety

### Treated Water Reuse

- . Green Belt Development
- . Landscaping & Gardening
- . Dust Suppression
- . Utility Water Supply

### PLC (Programmable Logic Controller)

- . Automatic pump operation
- . Protects equipment
- . Controls process

### SCADA System

- . Real-time monitoring
- . Indicates pump status
- . Monitors tank levels
- . Monitors flow rate



# Instrumentation, Control & Performance Analysis

The VPA STP employs a comprehensive instrumentation and SCADA-based monitoring system to ensure real-time process control, regulatory compliance, and early fault detection across all treatment stages.



## Flow Meters

Electromagnetic flow meters measure influent, RAS, WAS, and effluent flows. Critical for mass balance and hydraulic loading calculations.



## pH Meter

Continuous inline pH monitoring at inlet and outlet. Maintains optimal pH (6.5–8.5) for biological activity and discharge compliance.



## DO Meter

Dissolved Oxygen probes in aeration tank maintain DO at 2–3 mg/L. Controls blower operation via PLC for energy efficiency.



## Turbidity Meter

Measures effluent clarity (NTU). Ensures secondary clarifier performance and final water quality before discharge or reuse.



## Level Sensors

Ultrasonic and float-type level sensors in sumps, tanks, and clarifiers. Enable automated pump control and overflow prevention.



## SCADA System

Supervisory Control and Data Acquisition system provides centralised monitoring, data logging, alarm management, and remote control of all equipment.

## 9.7 Performance Analysis – Effluent Quality vs. CPCB Standards

| Parameter      | Raw Sewage | Treated Effluent | CPCB Limit | Removal % |
|----------------|------------|------------------|------------|-----------|
| BOD (mg/L)     | 300        | < 20             | 30         | 93%       |
| COD (mg/L)     | 600        | < 50             | 100        | 92%       |
| TSS (mg/L)     | 350        | < 30             | 50         | 91%       |
| pH             | 6.0–7.0    | 7.0–8.0          | 5.5–9.0    | —         |
| Fecal Coliform | High       | < 100/100ml      | 100        | >99%      |

# DELTA POND

A storage pond used to collect treated water before reuse or discharge. It acts as a balancing reservoir and provides additional settling of fine particles. Water from the delta pond is often used for irrigation, gardening, or industrial purposes.



# Safety, Environmental Benefits & Conclusion

## 10.1 Safety Measures at VPA STP

STP operations involve biological, chemical, and electrical hazards. Strict safety protocols are enforced:

|                                                                                                                                                                                                                             |                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>PPE Requirements</b><br>Safety helmet, rubber gloves, safety shoes, goggles, and respirator masks mandatory in all process areas.   | <br><b>Electrical Safety</b><br>LOTO (Lockout-Tagout) procedures for all electrical maintenance. RCD protection on all circuits.        |
| <br><b>Chemical Safety</b><br>Chlorine handling with proper ventilation, gas detectors, and emergency showers. MSDS displayed prominently. | <br><b>Biological Safety</b><br>Vaccination for Hepatitis A & Typhoid for all operators. Handwashing stations at all entry/exit points. |

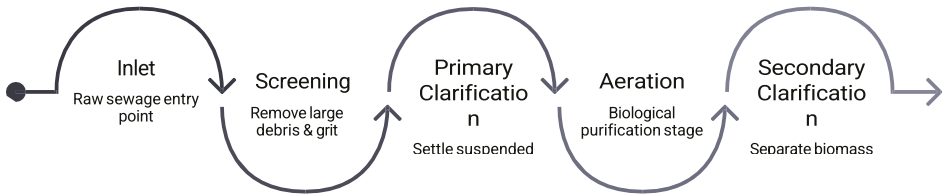
## 10.2 Environmental Benefits

- **Pollution Control:** Prevents untreated sewage from entering Bay of Bengal
- **Groundwater Protection:** Eliminates seepage contamination risk
- **Marine Ecosystem:** Protects fish breeding grounds and coral habitats
- **Water Conservation:** 10 MLD treated water reused daily, reducing freshwater demand
- **Sustainable Port Operations:** Aligns with Sagarmala and Green Port initiatives

## 10.3 Major Mechanical Components Summary

| Equipment         | Primary Function     |
|-------------------|----------------------|
| Bar Screen        | Remove large debris  |
| Grit Chamber      | Remove sand & grit   |
| Sewage Pumps      | Transfer sewage      |
| Primary Clarifier | Sedimentation        |
| Air Blowers       | Oxygen supply        |
| Diffusers         | Air distribution     |
| Chlorinator       | Disinfection         |
| Sludge Thickener  | Sludge concentration |

## 10.4 Conclusion – Journey of Sewage to Reuse



The VPA Sewage Treatment Plant stands as a model of operational excellence in port environmental management. With **93% BOD removal**, **92% COD reduction**, and complete pathogen elimination, the plant ensures zero marine pollution while enabling sustainable water reuse. The integration of advanced mechanical systems, biological treatment, and SCADA-based monitoring positions VPA STP as a benchmark for industrial wastewater management in India.

**93%**  
BOD Removal  
Exceeds CPCB standards

**>99%**  
Pathogen Kill  
Through chlorination

**10MLD**  
Water Reused  
Daily non-potable reuse

**0**  
Marine Discharge  
Untreated sewage